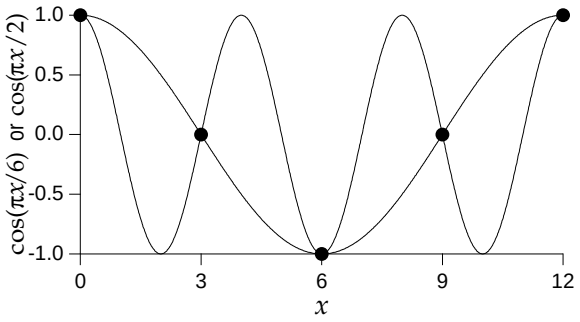

Figure 2.9 Aliasing and the Nyquist frequency. The two curves are the functions $\cos(\pi x/6)$ and $\cos(\pi x/2)$ plotted against x, and the dots show points sampled with a spacing of $\Delta x = 3$. The Nyquist frequency in this case is $\pi/3$, and the two cosine curves match at the sampled points because their spatial frequencies satisfy the relation $2\pi/3 - \pi/6 = \pi/2$.

can be confused with one another in this way, a phenomenon known as aliasing. Conversely, if an image is constructed solely of frequencies less than K_{nyq}, it can be reconstructed perfectly from the finite set of samples provided by the array. There are 120 cones per degree at the fovea of the macaque retina, which makes $K_{\text{nyq}}/(2\pi) = 1/(2\Delta x) = 60$ cycles per degree. In this result, we have divided the right side of equation 2.20, which gives K_{nyq} in units of radians per degree, by 2π to convert the answer to cycles per degree.

2.4 Reverse-Correlation Methods: Simple Cells

The spike-triggered average for visual stimuli is defined, as in chapter 1, as the average over trials of stimuli evaluated at times $t_i - \tau$, where t_i for $i = 1, 2, \ldots, n$ are the spike times. Because the light intensity of a visual image depends on location as well as time, the spike-triggered average stimulus is a function of three variables,

$$C(x, y, \tau) = \frac{1}{\langle n \rangle} \left\langle \sum_{i=1}^{n} s(x, y, t_i - \tau) \right\rangle . \tag{2.21}$$

Here, as in chapter 1, the brackets denote trial averaging, and we have used the approximation $1/n \approx 1/\langle n \rangle$. $C(x, y, \tau)$ is the average value of the visual stimulus at the point (x, y) a time τ before a spike was fired. Similarly, we can define the correlation function between the firing rate at time t and the stimulus at time $t + \tau$, for trials of duration T, as

$$Q_{\text{rs}}(x, y, \tau) = \frac{1}{T} \int_0^T dt\, \mathsf{r}(t) s(x, y, t + \tau) . \tag{2.22}$$

The spike-triggered average is related to the reverse-correlation function, as discussed in chapter 1, by

$$C(x, y, \tau) = \frac{Q_{\mathrm{rs}}(x, y, -\tau)}{\langle r \rangle}, \tag{2.23}$$

where $\langle r \rangle$ is, as usual, the average firing rate over the entire trial, $\langle r \rangle = \langle n \rangle / T$.

To estimate the firing rate of a neuron in response to a particular image, we add a function of the output of a linear filter of the stimulus to the background firing rate r_0, as in equation 2.8, $\mathrm{r}_{\mathrm{est}}(t) = \mathrm{r}_0 + F(L(t))$. As in equation 2.7, the linear estimate $L(t)$ is obtained by integrating over the past history of the stimulus with a kernel acting as the weighting function. Because visual stimuli depend on spatial location, we must decide how contributions from different image locations are to be combined to determine $L(t)$. The simplest assumption is that the contributions from different spatial points sum linearly, so that $L(t)$ is obtained by integrating over all x and y values:

linear response estimate

$$L(t) = \int_0^\infty d\tau \int dx dy \, D(x, y, \tau) s(x, y, t - \tau). \tag{2.24}$$

The kernel $D(x, y, \tau)$ determines how strongly, and with what sign, the visual stimulus at the point (x, y) and at time $t - \tau$ affects the firing rate of the neuron at time t. As in equation 2.6, the optimal kernel is given in terms of the firing rate-stimulus correlation function, or the spike-triggered average, for a white-noise stimulus with variance parameter σ_s^2 by

$$D(x, y, \tau) = \frac{Q_{\mathrm{rs}}(x, y, -\tau)}{\sigma_s^2} = \frac{\langle r \rangle C(x, y, \tau)}{\sigma_s^2}. \tag{2.25}$$

The kernel $D(x, y, \tau)$ defines the space-time receptive field of a neuron. Because $D(x, y, \tau)$ is a function of three variables, it can be difficult to measure and visualize. For some neurons, the kernel can be written as a product of two functions, one that describes the spatial receptive field and the other, the temporal receptive field,

space-time receptive field

$$D(x, y, \tau) = D_{\mathrm{s}}(x, y) D_{\mathrm{t}}(\tau). \tag{2.26}$$

Such neurons are said to have separable space-time receptive fields. Separability requires that the spatial structure of the receptive field not change over time except by an overall multiplicative factor. When $D(x, y, \tau)$ cannot be written as the product of two terms, the neuron is said to have a nonseparable space-time receptive field. Given the freedom in equation 2.8 to set the scale of D (by suitably adjusting the function F), we typically normalize D_{s} so that its integral is 1, and use a similar rule for the components from which D_{t} is constructed. We begin our analysis by studying first the spatial and then the temporal components of a separable space-time receptive field, and then proceed to the nonseparable case. For simplicity, we ignore the possibility that cells can have slightly different receptive fields for the two eyes, which underlies the disparity tuning considered in chapter 1.

separable receptive field

nonseparable receptive field

Spatial Receptive Fields

Figures 2.10A and C show the spatial structure of spike-triggered average stimuli for two simple cells in the primary visual cortex of a cat (area 17) with approximately separable space-time receptive fields. These receptive fields are elongated in one direction. There are some regions within the receptive field where D_s is positive, called ON regions, and others where it is negative, called OFF regions. The integral of the linear kernel times the stimulus can be visualized by noting how the OFF (black) and ON (white) regions overlap the image (see figure 2.11) . The response of a neuron is enhanced if ON regions are illuminated ($s > 0$) or if OFF regions are darkened ($s < 0$) relative to the background level of illumination. Conversely, they are suppressed by darkening ON regions or illuminating OFF regions. As a result, the neurons of figures 2.10A and C respond most vigorously to light-dark edges positioned along the border between the ON and OFF regions, and oriented parallel to this border and to the elongated direction of the receptive fields (figure 2.11). Figures 2.10 and 2.11 show receptive fields with two major subregions. Simple cells are found with from one to five subregions. Along with the ON-OFF patterns we have seen, another typical arrangement is a three-lobed receptive field with OFF-ON-OFF or ON-OFF-ON subregions.

Gabor function

A mathematical approximation of the spatial receptive field of a simple cell is provided by a Gabor function, which is a product of a Gaussian function and a sinusoidal function. Gabor functions are by no means the only functions used to fit spatial receptive fields of simple cells. For example, gradients of Gaussians are sometimes used. However, we will stick to Gabor functions, and to simplify the notation, we choose the coordinates x and y so that the borders between the ON and OFF regions are parallel to the y axis. We also place the origin of the coordinates at the center of the receptive field. With these choices, we can approximate the observed receptive field structures using the Gabor function

$$D_s(x, y) = \frac{1}{2\pi\sigma_x\sigma_y} \exp\left(-\frac{x^2}{2\sigma_x^2} - \frac{y^2}{2\sigma_y^2}\right)\cos(kx - \phi)\,. \tag{2.27}$$

rf size σ_x, σ_y

preferred spatial frequency k

preferred spatial phase ϕ

The parameters in this function determine the properties of the spatial receptive field: σ_x and σ_y determine its extent in the x and y directions, respectively; k, the preferred spatial frequency, determines the spacing of light and dark bars that produce the maximum response (the preferred spatial wavelength is $2\pi/k$); and ϕ is the preferred spatial phase, which determines where the ON-OFF boundaries fall within the receptive field. For this spatial receptive field, the sinusoidal grating described by equation 2.18 that produces the maximum response for a fixed value of A has $K = k$, $\Phi = \phi$, and $\Theta = 0$.

Figures 2.10B and 2.10D show Gabor functions chosen specifically to match the data in figures 2.10A and 2.10C. Figure 2.12 shows x and y plots of a variety of Gabor functions with different parameter values. As

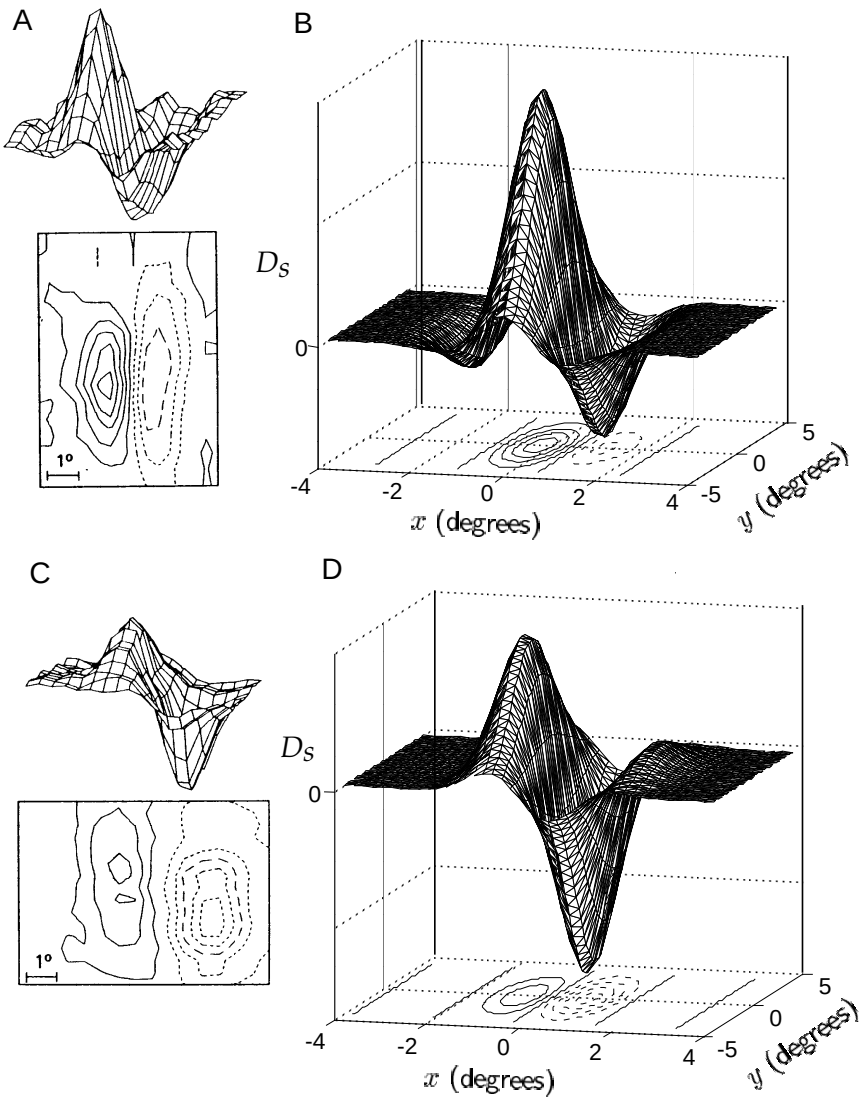

Figure 2.10 Spatial receptive field structure of simple cells. (A) and (C) Spatial structure of the receptive fields of two neurons in cat primary visual cortex determined by averaging stimuli between 50 ms and 100 ms prior to an action potential. The upper plots are three-dimensional representations, with the horizontal dimensions acting as the x-y plane and the vertical dimension indicating the magnitude and sign of $D_s(x, y)$. The lower contour plots represent the x-y plane. Regions with solid contour curves are ON areas where $D_s(x, y) > 0$, and regions with dashed contours are OFF areas where $D_s(x, y) < 0$. (B) and (D) Gabor functions (equation 2.27) with $\sigma_x = 1^\circ$, $\sigma_y = 2^\circ$, $1/k = 0.56^\circ$, and $\phi = 1 - \pi/2$ (B) or $\phi = 1 - \pi$ (D), chosen to match the receptive fields in A and C. (A and C adapted from Jones and Palmer, 1987a.)

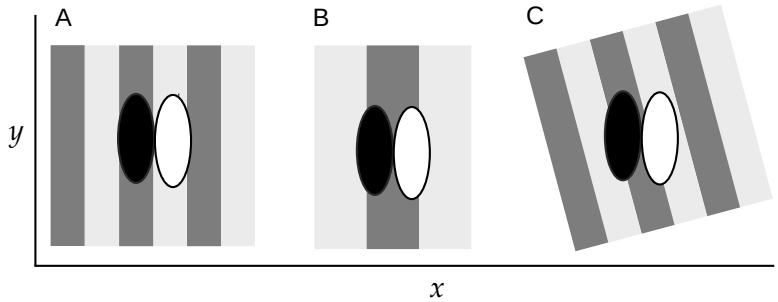

Figure 2.11 Grating stimuli superimposed on spatial receptive fields similar to those shown in figure 2.10. The receptive field is shown as two oval regions, one dark to represent an OFF area where $D_s < 0$ and one white to denote an ON region where $D_s > 0$. (A) A grating with the spatial wavelength, orientation, and spatial phase shown produces a high firing rate because a dark band completely overlaps the OFF area of the receptive field and a light band overlaps the ON area. (B) The grating shown is nonoptimal due to a mismatch in both the spatial phase and frequency, so that the ON and OFF regions each overlap both light and dark stripes. (C) The grating shown is at a nonoptimal orientation because each region of the receptive field overlaps both light and dark stripes.

seen in figure 2.12, Gabor functions can have various types of symmetry, and variable numbers of significant oscillations (or subregions) within the Gaussian envelope. The number of subregions within the receptive field is determined by the product $k\sigma_x$ and is typically expressed in terms of a quantity known as the bandwidth b. The bandwidth is defined as $b = \log_2(K_+/K_-)$, where $K_+ > k$ and $K_- < k$ are the spatial frequencies of gratings that produce one-half the response amplitude of a grating with $K = k$. High bandwidths correspond to low values of $k\sigma_x$, meaning that the receptive field has few subregions and poor spatial frequency selectivity. Neurons with more subfields are more selective to spatial frequency, and they have smaller bandwidths and larger values of $k\sigma_x$.

bandwidth

The bandwidth is the width of the spatial frequency tuning curve measured in octaves. The spatial frequency tuning curve as a function of K for a Gabor receptive field with preferred spatial frequency k and receptive field width σ_x is proportional to $\exp(-\sigma_x^2(k-K)^2/2)$ (see equation 2.34 below). The values of K_+ and K_- needed to compute the bandwidth are thus determined by the condition $\exp(-\sigma_x^2(k-K_\pm)^2/2) = 1/2$. Solving this equation gives $K_\pm = k \pm (2\ln(2))^{1/2}/\sigma_x$, from which we obtain

$$b = \log_2\left(\frac{k\sigma_x + \sqrt{2\ln(2)}}{k\sigma_x - \sqrt{2\ln(2)}}\right) \quad \text{or} \quad k\sigma_x = \sqrt{2\ln(2)}\,\frac{2^b+1}{2^b-1}\,. \tag{2.28}$$

Bandwidth is defined only if $k\sigma_x > (2\ln(2))^{1/2}$, but this is usually the case. Bandwidths typically range from about 0.5 to 2.5, corresponding to $k\sigma_x$ between 1.7 and 6.9.

The response characterized by equation 2.27 is maximal if light-dark edges are parallel to the y axis, so the preferred orientation angle is 0. An arbitrary preferred orientation, θ, can be created by rotating the coordinates,

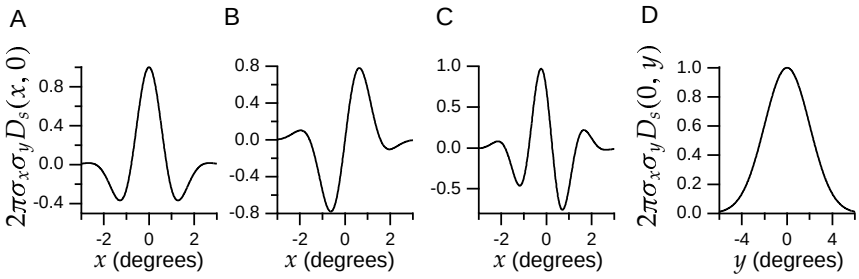

Figure 2.12 Gabor functions of the form given by equation 2.27. For convenience we plot the dimensionless function $2\pi\sigma_x\sigma_y D_s$. (A) A Gabor function with $\sigma_x = 1^\circ$, $1/k = 0.5^\circ$, and $\phi = 0$ plotted as a function of x for $y = 0$. This function is symmetric about $x = 0$. (B) A Gabor function with $\sigma_x = 1^\circ$, $1/k = 0.5^\circ$, and $\phi = \pi/2$ plotted as a function of x for $y = 0$. This function is antisymmetric about $x = 0$ and corresponds to using a sine instead of a cosine function in equation 2.27. (C) A Gabor function with $\sigma_x = 1^\circ$, $1/k = 0.33^\circ$, and $\phi = \pi/4$ plotted as a function of x for $y = 0$. This function has no particular symmetry properties with respect to $x = 0$. (D) The Gabor function of equation 2.27 with $\sigma_y = 2^\circ$ plotted as a function of y for $x = 0$. This function is simply a Gaussian.

making the substitutions $x \to x\cos(\theta) + y\sin(\theta)$ and $y \to y\cos(\theta) - x\sin(\theta)$ in equation 2.27. This produces a spatial receptive field that is maximally responsive to a grating with $\Theta = \theta$. Similarly, a receptive field centered at the point (x_0, y_0) rather than at the origin can be constructed by making the substitutions $x \to x - x_0$ and $y \to y - y_0$.

preferred orientation θ

rf center x_0, y_0

Temporal Receptive Fields

Figure 2.13 reveals the temporal development of the space-time receptive field of a neuron in the cat primary visual cortex through a series of snapshots of its spatial receptive field. More than 300 ms prior to a spike, there is little correlation between the visual stimulus and the upcoming spike. Around 210 ms before the spike ($\tau = 210$ ms), a two-lobed OFF-ON receptive field, similar to the ones in figure 2.10, is evident. As τ decreases (recall that τ measures time in a reversed sense), this structure first fades away and then reverses, so that the receptive field 75 ms before a spike has the opposite sign from what appeared at $\tau = 210$ ms. Due to latency effects, the spatial structure of the receptive field is less significant for $\tau < 75$ ms. The stimulus preferred by this cell is thus an appropriately aligned dark-light boundary that reverses to a light-dark boundary over time.

Reversal effects like those seen in figure 2.13 are a common feature of space-time receptive fields. Although the magnitudes and signs of the different spatial regions in figure 2.13 vary over time, their locations and shapes remain fairly constant. This indicates that the neuron has, to a good approximation, a separable space-time receptive field. When a space-time receptive field is separable, the reversal can be described by a function $D_t(\tau)$ that rises from 0, becomes positive, then negative, and ultimately

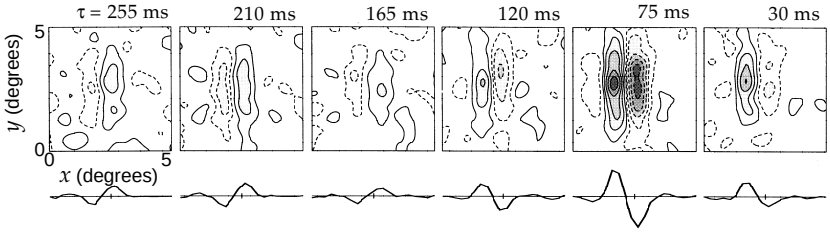

Figure 2.13 Temporal evolution of a spatial receptive field. Each panel is a plot of $D(x, y, \tau)$ for a different value of τ. As in figure 2.10, regions with solid contour curves are areas where $D(x, y, \tau) > 0$ and regions with dashed contours have $D(x, y, \tau) < 0$. The curves below the contour diagrams are one-dimensional plots of the receptive field as a function of x alone. The receptive field is maximally different from 0 for $\tau = 75$ ms with the spatial receptive field reversed from what it was at $\tau = 210$ ms. (Adapted from DeAngelis et al., 1995.)

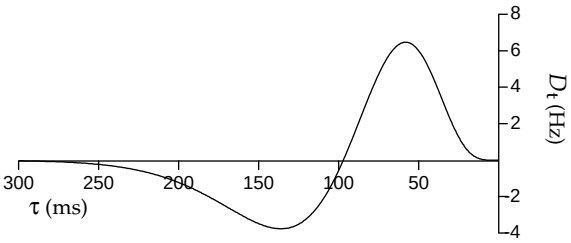

Figure 2.14 Temporal structure of a receptive field. The function $D_t(\tau)$ of equation 2.29 with $\alpha = 1/(15 \text{ ms})$.

returns to 0 as τ increases. Adelson and Bergen (1985) proposed the function shown in figure 2.14,

$$D_t(\tau) = \alpha \exp(-\alpha\tau) \left(\frac{(\alpha\tau)^5}{5!} - \frac{(\alpha\tau)^7}{7!} \right) , \tag{2.29}$$

for $\tau \geq 0$, and $D_t(\tau) = 0$ for $\tau < 0$. Here, α is a constant that sets the scale for the temporal development of the function. Single-phase responses are also seen for V1 neurons, and these can be described by eliminating the second term in equation 2.29. Three-phase responses, which are sometimes seen, must be described by a more complicated function.

Response of a Simple Cell to a Counterphase Grating

The response of a simple cell to a counterphase grating stimulus (equation 2.18) can be estimated by computing the function $L(t)$. For the separable receptive field given by the product of the spatial factor in equation 2.27 and the temporal factor in 2.29, the linear estimate of the response can be written as the product of two terms,

$$L(t) = L_s L_t(t) , \tag{2.30}$$

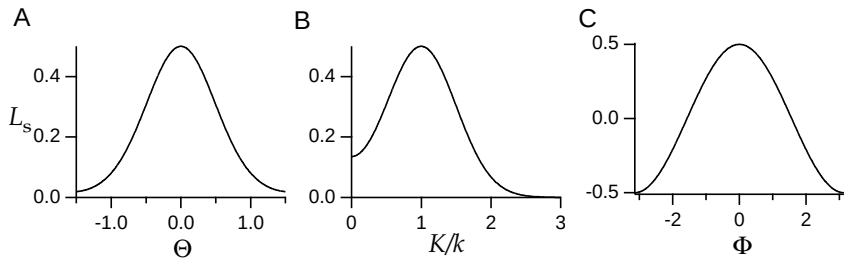

Figure 2.15 Selectivity of a Gabor filter with $\theta = \phi = 0$, $\sigma_x = \sigma_y = \sigma$, and $k\sigma = 2$ acting on a cosine grating with $A = 1$. (A) L_s as a function of stimulus orientation Θ for a grating with the preferred spatial frequency and phase, $K = k$ and $\Phi = 0$. (B) L_s as a function of the ratio of the stimulus spatial frequency to its preferred value, K/k, for a grating oriented in the preferred direction $\Theta = 0$ and with the preferred phase $\Phi = 0$. (C) L_s as a function of stimulus spatial phase Φ for a grating with the preferred spatial frequency and orientation, $K = k$ and $\Theta = 0$.

where

$$L_s = \int dxdy\, D_s(x, y) A \cos\left(Kx\cos(\Theta) + Ky\sin(\Theta) - \Phi\right) \tag{2.31}$$

and

$$L_t(t) = \int_0^\infty d\tau\, D_t(\tau) \cos\left(\omega(t - \tau)\right) . \tag{2.32}$$

The reader is invited to compute these integrals for the case $\sigma_x = \sigma_y = \sigma$. To show the selectivity of the resulting spatial receptive fields, we plot (in figure 2.15) L_s as functions of the parameters Θ, K, and Φ that determine the orientation, spatial frequency, and spatial phase of the stimulus. It is also instructive to write out L_s for various special parameter values. First, if the spatial phase of the stimulus and the preferred spatial phase of the receptive field are 0 ($\Phi = \phi = 0$), we find that

$$L_s = A \exp\left(-\frac{\sigma^2(k^2 + K^2)}{2}\right) \cosh\left(\sigma^2 kK\cos(\Theta)\right) , \tag{2.33}$$

which determines the orientation and spatial frequency tuning for an optimal spatial phase. Second, for a grating with the preferred orientation $\Theta = 0$ and a spatial frequency that is not too small, the full expression for L_s can be simplified by noting that $\exp(-\sigma^2 kK) \approx 0$ for the values of $k\sigma$ normally encountered (for example, if $K = k$ and $k\sigma = 2$, $\exp(-\sigma^2 kK) = 0.02$). Using this approximation, we find

$$L_s = \frac{A}{2} \exp\left(-\frac{\sigma^2(k - K)^2}{2}\right) \cos(\phi - \Phi) , \tag{2.34}$$

which reveals a Gaussian dependence on spatial frequency and a cosine dependence on spatial phase.

The amplitude of the sinusoidally oscillating linear response estimate (equation 2.32) is plotted as a function of the temporal frequency of the

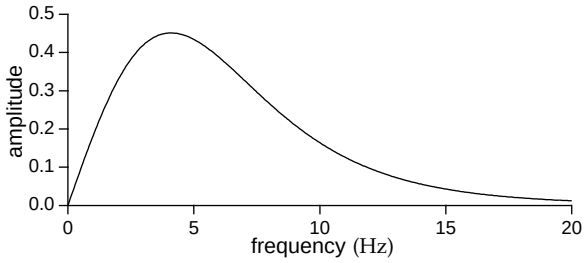

Figure 2.16 Frequency response of a model simple cell based on the temporal kernel of equation 2.29. The amplitude of the sinusoidal oscillations of $L_t(t)$ produced by a counterphase grating is plotted as a function of the temporal oscillation frequency, $\omega/2\pi$.

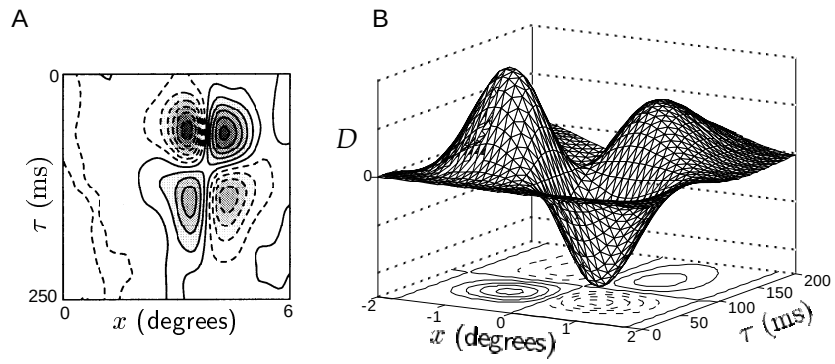

Figure 2.17 A separable space-time receptive field. (A) An x-τ plot of an approximately separable space-time receptive field from cat primary visual cortex. OFF regions are shown with dashed contour lines and ON regions with solid contour lines. The receptive field has side-by-side OFF and ON regions that reverse as a function of τ. (B) Mathematical description of the space-time receptive field in A constructed by multiplying a Gabor function (evaluated at $y = 0$) with $\sigma_x = 1°$, $1/k = 0.56°$, and $\phi = \pi/2$ by the temporal kernel of equation 2.29 with $1/\alpha = 15$ ms. (A adapted from DeAngelis et al., 1995.)

stimulus ($\omega/2\pi$ rather than the angular frequency ω) in figure 2.16. The peak value around 4 Hz and roll-off above 10 Hz are typical for V1 neurons and for cortical neurons in other primary sensory areas as well.

Space-Time Receptive Fields

To display the function $D(x, y, \tau)$ in a space-time plot rather than as a sequence of spatial plots (as in figure 2.13), we suppress the y dependence and plot an x-τ projection of the space-time kernel. Figure 2.17A shows a space-time plot of the receptive field of a simple cell in the cat primary visual cortex. This receptive field is approximately separable, and it has OFF and ON subregions that reverse to ON and OFF subregions as a function of τ, similar to the reversal seen in figure 2.13. Figure 2.17B shows an

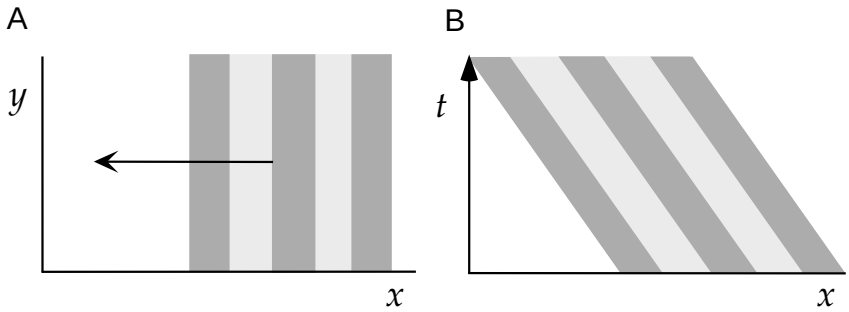

Figure 2.18 Space and space-time diagrams of a moving grating. (A) A vertically oriented grating moves to the left on a two-dimensional screen. (B) The space-time diagram of the image in A. The x location of the dark and light bands moves to the left as time progresses upward, representing the motion of the grating.

x-τ plot of a separable space-time kernel, similar to the one in figure 2.17A, generated by multiplying a Gabor function by the temporal kernel of equation 2.29.

We can also plot the visual stimulus in a space-time diagram, suppressing the y coordinate by assuming that the image does not vary as a function of y. For example, figure 2.18A shows a grating of vertically oriented stripes moving to the left on an x-y plot. In the x-t plot of figure 2.18B, this image appears as a series of sloped dark and light bands. These represent the projection of the image in figure 2.18A onto the x axis evolving as a function of time. The leftward slope of the bands corresponds to the leftward movement of the image.

Most neurons in primary visual cortex do not respond strongly to static images, but respond vigorously to flashed and moving bars and gratings. The receptive field structure of figure 2.17 reveals why this is the case, as is shown in figures 2.19 and 2.20. The image in figures 2.19A-C is a dark bar that is flashed on for a brief period of time. To describe the linear response estimate at different times, we consider a space-time receptive field similar to the one in figure 2.17A. The receptive field is positioned at three different times in figures 2.19A, B, and C. The height of the horizontal axis of the receptive field diagram indicates the time when the estimation is being made. Figure 2.19A corresponds to an estimate of $L(t)$ at the moment when the image first appears. At this time, $L(t) = 0$. As time progresses, the receptive field diagram moves upward. Figure 2.19B generates an estimate at the moment of maximum response, when the dark image overlaps the OFF area of the space-time receptive field, producing a positive contribution to $L(t)$. Figure 2.19C shows a later time when the dark image overlaps an ON region, generating a negative $L(t)$. The response for this flashed image is thus transient firing followed by suppression, as shown in Figure 2.19D.

Figures 2.19E and 2.19F show why a static dark bar is an ineffective stimulus. The static bar overlaps both the OFF region for small τ and the re-

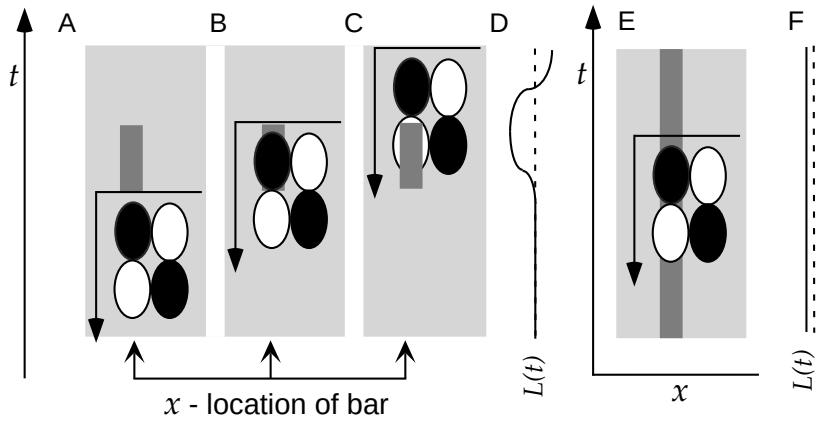

Figure 2.19 Responses to dark bars estimated from a separable space-time receptive field. Dark ovals in the receptive field diagrams are OFF regions and light circles are ON regions. The linear estimate of the response at any time is determined by positioning the receptive field diagram so that its horizontal axis matches the time of response estimation and noting how the OFF and ON regions overlap with the image. (A-C) The image is a dark bar that is flashed on for a short interval of time. There is no response (A) until the dark image overlaps the OFF region (B) when $L(t) > 0$. The response is later suppressed when the dark bar overlaps the ON region (C) and $L(t) < 0$. (D) A plot of $L(t)$ versus time corresponding to the responses generated in A-C. Time runs vertically in this plot, and $L(t)$ is plotted horizontally with the dashed line indicating the zero axis and positive values plotted to the left. (E) The image is a static dark bar. The bar overlaps both an OFF and an ON region, generating opposing positive and negative contributions to $L(t)$. (F) The weak response corresponding to E, plotted as in D.

versed ON region for large τ, generating opposing positive and negative contributions to $L(t)$. The flashed dark bar of figures 2.19A-C is a more effective stimulus because there is a time when it overlaps only the OFF region.

Figure 2.20 shows why a moving grating is a particularly effective stimulus. The grating moves to the left in 2.20A-C. At the time corresponding to the positioning of the receptive field diagram in 2.20A, a dark band stimulus overlaps both OFF regions and light bands overlap both ON regions. Thus, all four regions contribute positive amounts to $L(t)$. As time progresses and the receptive field moves upward in the figure, the alignment will sometimes be optimal, as in 2.20A, and sometimes nonoptimal, as in 2.20B. This produces an $L(t)$ that oscillates as a function of time between positive and negative values (2.20C). Figures 2.20D-F show that a neuron with this receptive field responds equally to a grating moving to the right. Like the left-moving grating in figures 2.20A-C, the right-moving grating can overlap the receptive field in an optimal manner (2.20D), producing a strong response, or in a maximally negative manner (2.20E), producing strong suppression of response, again resulting in an oscillating response (2.20F). Separable space-time receptive fields can produce responses that are maximal for certain speeds of grating motion, but they are not sensitive to the direction of motion.

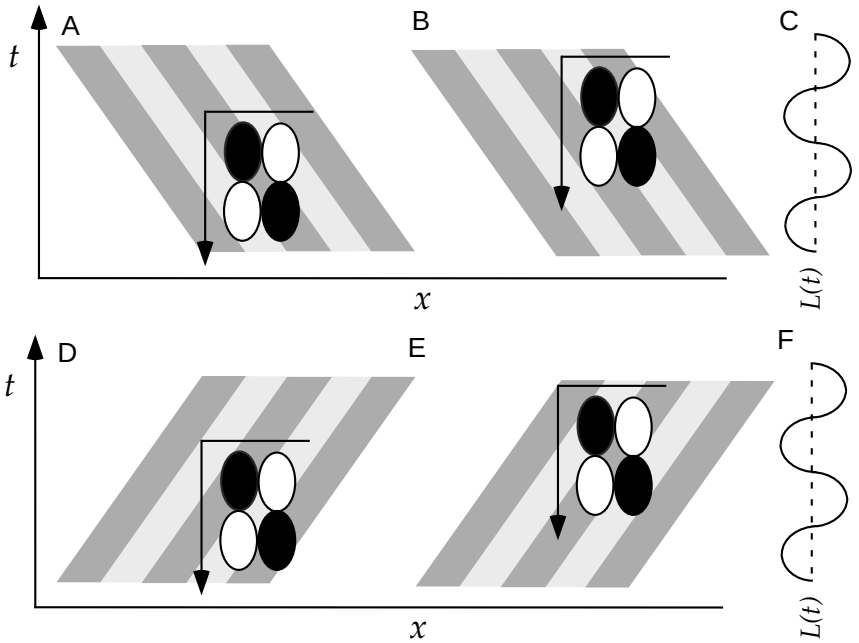

Figure 2.20 Responses to moving gratings estimated from a separable space-time receptive field. The receptive field is the same as in figure 2.19. (A-C) The stimulus is a grating moving to the left. At the time corresponding to A, OFF regions overlap with dark bands and ON regions with light bands, generating a strong response. At the time of the estimate in B, the alignment is reversed, and $L(t)$ is negative. (C) A plot of $L(t)$ versus time corresponding to the responses generated in A-B. Time runs vertically in this plot and $L(t)$ is plotted horizontally, with the dashed line indicating the zero axis and positive values plotted to the left. (D-F) The stimulus is a grating moving to the right. The responses are identical to those in A-C.

Nonseparable Receptive Fields

Many neurons in primary visual cortex are selective for the direction of motion of an image. Accounting for direction selectivity requires nonseparable space-time receptive fields. An example of a nonseparable receptive field is shown in figure 2.21A. This neuron has a three-lobed OFF-ON-OFF spatial receptive field, and these subregions shift to the left as time moves forward (and τ decreases). This means that the optimal stimulus for this neuron has light and dark areas that move toward the left. One way to describe a nonseparable receptive field structure is to use a separable function constructed from a product of a Gabor function for D_s and equation 2.29 for D_t, but to write these as functions of a mixture or rotation of the x and τ variables. The rotation of the space-time receptive field, as seen in figure 2.21B, is achieved by mixing the space and time coordinates, using the transformation

$$D(x, y, \tau) = D_s(x', y) D_t(\tau') \tag{2.35}$$

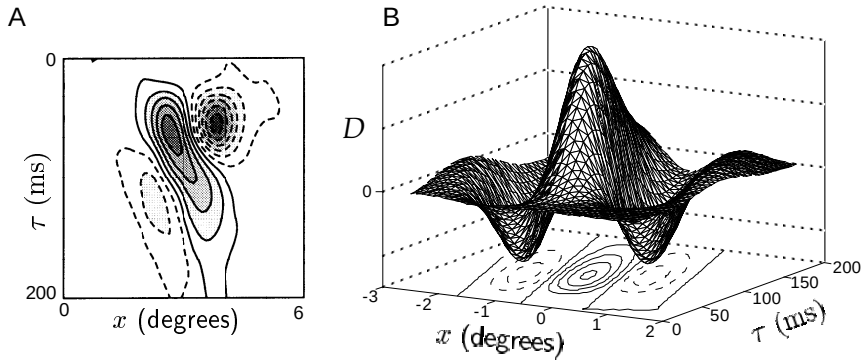

Figure 2.21 A nonseparable space-time receptive field. (A) An x-τ plot of the space-time receptive field of a neuron from cat primary visual cortex. OFF regions are shown with dashed contour lines and ON regions with solid contour lines. The receptive field has a central ON region and two flanking OFF regions that shift to the left over time. (B) Mathematical description of the space-time receptive field in A constructed from equations 2.35 - 2.37. The Gabor function used (evaluated at $y = 0$) had $\sigma_x = 1^\circ$, $1/k = 0.5^\circ$, and $\phi = 0$. D_t is given by the expression in equation 2.29 with $\alpha = 20$ ms, except that the second term, with the seventh power function, was omitted because the receptive field does not reverse sign in this example. The x-τ rotation angle used was $\psi = \pi/9$, and the conversion factor was $c = 0.02\ ^\circ/\text{ms}$. (A adapted from DeAngelis et al., 1995.)

with

$$x' = x\cos(\psi) - c\tau\sin(\psi) \tag{2.36}$$

and

$$\tau' = \tau\cos(\psi) + \frac{x}{c}\sin(\psi)\,. \tag{2.37}$$

The factor c converts between the units of time (ms) and space (degrees), and ψ is the space-time rotation angle. The rotation operation is not the only way to generate nonseparable space-time receptive fields. They are often constructed by adding together two or more separable space-time receptive fields with different spatial and temporal characteristics.

Figure 2.22 shows how a nonseparable space-time receptive field can produce a response that is sensitive to the direction of motion of a grating. Figures 2.22A-C show a left-moving grating and, in 2.22A, the receptive field is positioned at a time when a light area of the image overlaps the central ON region and dark areas overlap the flanking OFF regions. This produces a large positive $L(t)$. At other times, the alignment is nonoptimal (2.22B), and over time, $L(t)$ oscillates between large positive and negative values (2.22C). The nonseparable space-time receptive field does not overlap optimally with the right-moving grating of figures 2.22D-F at any time, and the response is correspondingly weaker (2.22F). Thus, a neuron with a nonseparable space-time receptive field can be selective for the direction of motion of a grating and for its velocity, responding most vigorously to an optimally spaced grating moving at a velocity given, in terms of the parameters in equation 2.36, by $c\tan(\psi)$.

direction selectivity

preferred velocity

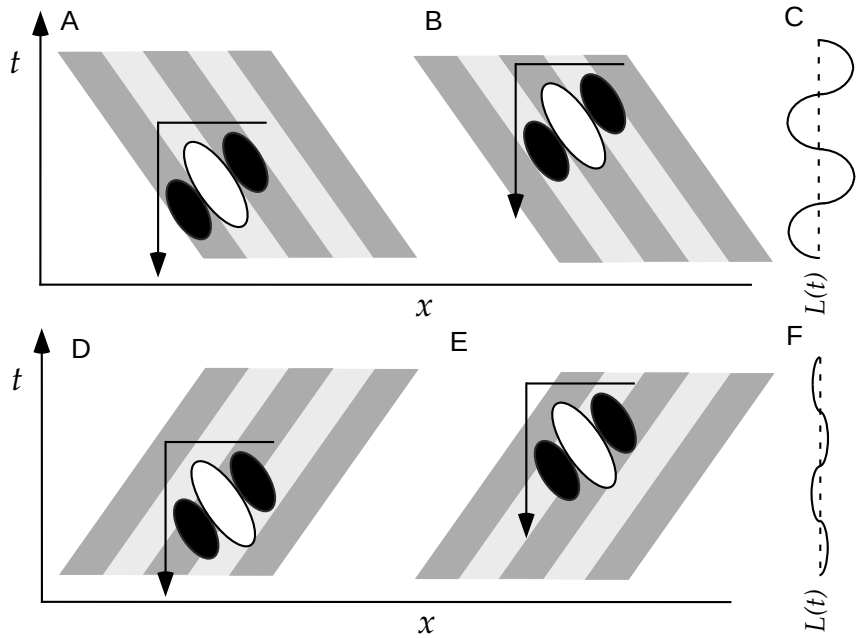

Figure 2.22 Responses to moving gratings estimated from a nonseparable space-time receptive field. Dark areas in the receptive field diagrams represent OFF regions and light areas, ON regions. (A-C) The stimulus is a grating moving to the left. At the time corresponding to A, OFF regions overlap with dark bands and the ON region overlaps a light band, generating a strong response. At the time of the estimate in B, the alignment is reversed, and $L(t)$ is negative. (C) A plot of $L(t)$ versus time corresponding to the responses generated in A and B. Time runs vertically in this plot, and $L(t)$ is plotted horizontally with the dashed line indicating the zero axis. (D-F) The stimulus is a grating moving to the right. Because of the tilt of the space-time receptive field, the alignment with the right-moving grating is never optimal and the response is weak (F).

Static Nonlinearities: Simple Cells

Once the linear response estimate $L(t)$ has been computed, the firing rate of a visually responsive neuron can be approximated by using equation 2.8, $\mathsf{r}_{\text{est}}(t) = \mathsf{r}_0 + F(L(t))$, where F is an appropriately chosen static nonlinearity. The simplest choice for F consistent with the positive nature of firing rates is rectification, $F = G[L]_+$, with G set to fit the magnitude of the measured firing rates. However, this choice makes the firing rate a linear function of the contrast amplitude, which does not match the data on the contrast dependence of visual responses. Neural responses saturate as the contrast of the image increases, and are more accurately described by $\mathsf{r} \propto A^n/(A_{1/2}^n + A^n)$ where n is near 2, and $A_{1/2}$ is a parameter equal to the contrast amplitude that produces a half-maximal response. This led Heeger (1992) to propose that an appropriate static nonlinearity to use is

contrast saturation

$$F(L) = \frac{G[L]_+^2}{A_{1/2}^2 + G[L]_+^2}\,, \tag{2.38}$$

because this reproduces the observed contrast dependence. A number of variants and extensions of this idea have also been considered, including, for example, that the denominator of this expression should include L factors for additional neurons with nearby receptive fields. This can account for the effects of visual stimuli outside the "classical" receptive field. Discussion of these effects is beyond the scope of this chapter.

2.5 Static Nonlinearities: Complex Cells

Recall that neurons in primary visual cortex are characterized as simple or complex. While linear methods, such as spike-triggered averages, are useful for revealing the properties of simple cells, at least to a first approximation, complex cells display features that are fundamentally incompatible with a linear description. The spatial receptive fields of complex cells cannot be divided into separate ON and OFF regions that sum linearly to generate the response. Areas where light and dark images excite the neuron overlap, making it difficult to measure and interpret spike-triggered average stimuli. Nevertheless, like simple cells, complex cells are selective to the spatial frequency and orientation of a grating. However, unlike simple cells, complex cells respond to bars of light or dark no matter where they are placed within the overall receptive field. Likewise, the responses of complex cells to grating stimuli show little dependence on spatial phase. Thus, a complex cell is selective for a particular type of image independent of its exact spatial position within the receptive field. This may represent an early stage in the visual processing that ultimately leads to position-invariant object recognition.

spatial-phase invariance

Complex cells also have temporal response characteristics that distinguish them from simple cells. Complex cell responses to moving gratings are approximately constant, not oscillatory as in figures 2.20 and 2.22. The firing rate of a complex cell responding to a counterphase grating oscillating with frequency ω has both a constant component and an oscillatory component with a frequency of 2ω, a phenomenon known as frequency doubling.

frequency doubling

Even though spike-triggered average stimuli and reverse-correlation functions fail to capture the response properties of complex cells, complex-cell responses can be described, to a first approximation, by a relatively straightforward extension of the reverse-correlation approach. The key observation comes from equation 2.34, which shows how the linear response estimate of a simple cell depends on spatial phase for an optimally oriented grating with K not too small. Consider two such responses, labeled L_1 and L_2, with preferred spatial phases ϕ and $\phi - \pi/2$. Including both the spatial and the temporal response factors, we find, for preferred spatial phase ϕ,

$$L_1 = AB(\omega, K)\cos(\phi - \Phi)\cos(\omega t - \delta)\,, \tag{2.39}$$